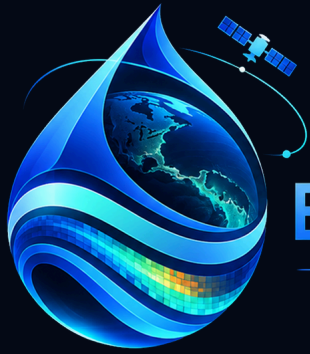




BLUEBAN 813
by Team Kanban

• ARAB YOUTH SPACE HACKATHON 2026 • 813 CHALLENGE

Early warning for the water in front of the UAE's intakes.

Satellites flag water that is unusual for that exact place and season. A person decides, a field team confirms, and the model learns, going live only when it provably improves.

Team Kanban

United Arab Emirates

Theme: Water Quality & Inland/Coastal Water Intelligence

Live: blueban813.kanbanstudios.ae • Code: github.com/4waiz/BlueBan-813



The UAE drinks the sea. When the sea turns, **the taps are at risk.**

~70

major desalination plants supply most of the UAE's drinking water, on its two coasts.

Source: UAE Government portal (u.ae), Water

4 months

the longest seawater reverse-osmosis shutdowns during the 2008–09 red tide in the Gulf of Oman, Fujairah's among them.

Source: IOC-UNESCO (2017), Harmful Algal Blooms and Desalination, Manuals and Guides 78

2–5 days

how often free Sentinel-2 images of this water arrive. Today they are looked at, not turned into decisions.

Source: ESA Sentinel-2 mission (Sentinel-2A, 2B and 2C)

Today the warning comes from the intake filters, a boat sample on a fixed calendar, or a complaint from the public: after the water has arrived.

Why now • Satellite 813 brings ~205 narrow bands built for water

Why now • a nine-year open archive lets each pixel be judged against its own past

Why now • cloud catalogues make screening the whole coast cheap

Built for the people who must **decide before the water reaches the intake.**

Desalination & power operators

Decide: step up pre-treatment, prepare the intake, send a boat?

Today: filters and lab results, after the fact.

Environment regulators & municipalities

Decide: where to sample today, what to report, when to warn a beach.

Today: a fixed sampling calendar and public reports.

Ports, fish farms & beaches

Decide: protect stock, pause works, close a beach.

Today: what can be seen from the shore.

• PROBLEM STATEMENT

We screen UAE coastal waters and the Shawka Dam reservoir for **unusual water colour**, on every Sentinel-2 pass since 2017, so that **desalination operators and environment regulators** can decide **whether, where and what to sample** before the water reaches an intake.

Boat samples on a fixed calendar



Extra sampling only where the evidence is, always with a clean-water control

"Is this normal?" answered from memory



Answered from years of images of that exact spot and season

A conclusion in an email



A decision with its evidence, model version and a tamper-evident log

Not a coloured map: **a loop that ends in a decision**, and learns from it.

WATCH

Sentinel-2 L2A over 10 UAE coastal areas and one inland dam, every pass.



DETECT

Each pixel against its **own** same-season history, not its neighbours.



DIAGNOSE

Colour fingerprint, Sentinel-3 same-morning check, Satellite 813 simulated.



VERIFY

A named analyst decides. A field team takes water samples.



ACT

Intake watch, alerts, a sampling plan with a clean-water control.



LEARN

Verified labels train a candidate that must pass a frozen test and a person's sign-off.

Satellites flag it. A person decides. The field confirms. **The model learns, and goes live only when it provably improves.**

Every decision lands in a tamper-evident (hash-chained) audit log, and every number on screen carries its source: scene ID, licence and algorithm version.

● DATA USED

Open data at the core, hyperspectral where it improves the result.

DATASET	PROVIDER · LEVEL	DATES	ROLE IN BLUEBAN
Sentinel-2 MSI L2A	ESA Copernicus via Planetary Computer · surface reflectance, 10–20 m	2017–2026 · 5,080 images	Detection, the primary sensor
Sentinel-3 OLCI WFR	EUMETSAT / ESA Copernicus · Level-2 water, 300 m	2017 – Feb 2026	Same-morning cross-check; 436 reference labels, never ground truth
Planet Tanager-1	Planet open archive (CC BY 4.0) · surface reflectance, >420 bands, 35 m	1 Jun 2025 scene	Real pixels → simulated 813 bands; the 813 test
EnMAP HSI L2A	DLR · L2A surface reflectance, 224 bands, 30 m	8 Sep 2022, 24 Apr 2024	Inland: Shawka Dam water and anomalies; 813 simulated on EnMAP
Landsat 8/9 + Sentinel-2	USGS, ESA · 30 m / 20 m	Sep 2023 – Oct 2025	Inland baseline, 28 images
ERA5 10 m wind	ECMWF / Copernicus C3S	Per event	Wind-only drift scenario (not an ocean forecast)
OpenStreetMap	OSM contributors (ODbL)	2026 extract	120 coastal assets, incl. 8 desalination and 22 power plants

SIMULATED Satellite 813 data was not available to teams in the PoC phase, so its published band set (~205 bands, 400–1700 nm, 20 m) is simulated from real hyperspectral pixels and labelled as simulated everywhere.

An **explainable baseline first**, then a model that has to earn its place.

1 · READ CORRECTLY

Sentinel-2 L2A with each product's BOA offset, SCL cloud and shadow masks, a glint-aware water mask, 20 m grid.



2 · WATER FEATURES

NDCI, MCI red-edge peak, hue angle, turbidity (generic formula). Shown as proxies, never as concentrations.



3 · AGAINST ITS OWN PAST

Robust z and seasonal percentile vs the same pixel ± 45 days in other years. Flag: $z \geq 3$, ≥ 95 th percentile, $\geq 0.5 \text{ km}^2$.



4 · DIAGNOSE

Event vs background spectrum, Sentinel-3 the same morning, simulated 813 bands.



5 · LEARN, SAFELY

L2 logistic candidate vs the rule on a frozen test set, bootstrap intervals, a person approves.

No leakage

Train and test split by area and month; 1.2 km spatial blocks for the 813 test.

Controls kept

Known false alarms stay in the data on purpose: the system must stand down.

Same code twice

Python and the in-browser trainer agree to $1e-7$; CI runs the whole loop.

Reproducible

A notebook reruns the Fujairah detection and the gate from committed sample data.

One screen: where the water is unusual, why, and what to do next.



- 1 The event, mapped. 17 Feb 2024 true colour and the anomaly off Fujairah, with a five-point sampling plan.
- 2 Evidence, then a decision. Proxies vs past years, Sentinel-3 agreement; Confirm, False alarm or Request a water sample.
- 3 Fingerprint and drift. The spectral cube, event-minus-normal spectrum and a wind-only drift scenario.

Live at blueban813.kanbanstudios.ae: review, sampling, retraining and gated promotion all run in the browser. The FastAPI backend runs the same loop on a database, with an HTTP API.

The question is **per intake**, not per incident.

ACTIVE MODEL
v1.0.0 - Live
rule-based image

LAST IMAGE
29 Sept 2026
07:03 UTC · Sentinel-2

NEXT PASS
in 10h 12m
Sentinel-2G · orbit 63

Judge Mode20:44:15 UTC
Tue, 06 Oct 2026

INCIDENT CONTROL

WATCH
Monitoring
10 areas

DETECT
Incident queue
2 open

DIAGNOSE
Colour fingerprint
52 × 813 km

VERIFY
Field samples
2 to review

ACT
Investigation
actions due

LEARN
Model improvement
0 new labels

OPERATIONS

INTAKE WATCH

ASSETS

EVIDENCE

VALIDATION

SATELLITE VIEW

INLAND · SHAWKA

SYSTEM

DATA & SOURCES

API

SETTINGS

ACT · PER INTAKE, NOT PER INCIDENT

INTAKE WATCH: IS THE WATER IN FRONT OF EACH INTAKE UNUSUAL?

1 Act now

4 Watch

25 Clear

1 Not monitored

STATUS	ASSET	NEAREST OPEN INCIDENT	DRIFT SCENARIO	EYES ON IT
ACT NOW	Kalba Power Plant Power plant (cooling water) · Fujairah / Qidfa / anchorage	12.1 km ENE BB-AE-2024-001 · 17 Feb 2024	9.81 km at +6 h	29 Sept 2026 next pass in 2d 10h
WATCH	Fujairah F2 power and desalination plant Desalination · Fujairah / Qidfa / anchorage	16.3 km SE BB-AE-2023-001 · 25 Oct 2023	15.16 km at +6 h	29 Sept 2026 next pass in 2d 10h
WATCH	Fujairah F3 power plant Power plant (cooling water) · Fujairah / Qidfa / anchorage	16.8 km SE BB-AE-2023-001 · 25 Oct 2023	15.69 km at +6 h	29 Sept 2026 next pass in 2d 10h
WATCH	Fujairah Qidfa F1 power and desalination plant Desalination · Fujairah / Qidfa / anchorage	17.3 km SE BB-AE-2023-001 · 25 Oct 2023	16.13 km at +6 h	29 Sept 2026 next pass in 2d 10h
WATCH	Khorfakkan Power Plant Power plant (cooling water) · Khor Fakkan	23.7 km SE BB-AE-2023-001 · 25 Oct 2023	22.61 km at +6 h	29 Sept 2026 next pass in 2d 10h
CLEAR	Al Zawra Power and Desalination Plant Desalination · Sharjah / Ajman	none within 30 km	-	29 Sept 2026 next pass in 2d 10h
CLEAR	Jebel Ali Power and Desalination Plant Desalination · Dubai / Jebel Ali	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Taweelah A1 IWPP Desalination · Al Taweelah / Khalifa Port	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Taweelah A2 IWPP Desalination · Al Taweelah / Khalifa Port	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Taweelah Reverse Osmosis Desalination · Al Taweelah / Khalifa Port	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Water desalination plant UAQ Desalination · Umm Al Quwain / Ras Al Khaimah	none within 30 km	-	29 Sept 2026 next pass in 2d 10h
CLEAR	Ajman Water & Electricity Power Plant Power plant (cooling water) · Sharjah / Ajman	none within 30 km	-	29 Sept 2026 next pass in 2d 10h
CLEAR	Emirates Global Aluminium Al Taweelah power plant Power plant (cooling water) · Al Taweelah / Khalifa Port	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Emirates Global Aluminium Jebel Ali power plant Power plant (cooling water) · Dubai / Jebel Ali	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Hamriyah Power Plant Power plant (cooling water) · Sharjah / Ajman	none within 30 km	-	29 Sept 2026 next pass in 2d 10h
CLEAR	JLT District Cooling Plant Power plant (cooling water) · Dubai / Jebel Ali	none within 30 km	-	29 Sept 2026 next pass in 10h 12m
CLEAR	Layyah Power Plant Power plant (cooling water) · Sharjah / Ajman	none within 30 km	-	29 Sept 2026 next pass in 2d 10h

Distances are to the facility centre (OpenStreetMap): intake positions are not public, so none is guessed. Status rules: Act now: an open incident within 15 km, or the drift scenario comes within 5 km · Watch: an open incident within 30 km.

WHAT THE OPERATOR RECEIVES

KALBA POWER PLANT

ACT NOW

An open incident within 15 km, or the drift scenario comes within 5 km

BLUEBAN 813 · ACT NOW · Kalba Power Plant

Unusual water 12.1 km to the ENE of the facility (possible algae bloom (high chlorophyll), 2.39 km²), seen by Sentinel-2 on 17 Feb 2024 · 06:49 UTC.
Unusual for this place and season: 100th percentile against 33 past same-season images.
Sentinel-3 cross-check: agrees.
Wind-only drift scenario: closest approach 9.81 km at +6 h (a scenario, not an ocean forecast).
Recommended: review the evidence and request water samples (the plan includes a clean-water control). Incident BB-AE-2024-001.
Not a confirmed harmful bloom; no concentration is claimed until water samples are tested.

Copy alert

Open the evidence

Plan water samples

HOW IT IS DELIVERED

The API raises an alert on each new incident and posts it as JSON to the operator's webhook (BLUEBAN_ALERT_WEBHOOK), from where their own email, SMS or control-room system takes over. This hosted demo shows the message instead of sending it.

Exposure score 0.034 (proximity × drift reach × sensitivity)

UAE COASTAL MONITORING

2 Active incidents

10 Areas monitored

18 Images (7 days)

0 New labels

Open data: Copernicus Sentinel-2/3 · Landsat · Planet Tanager (CC BY 4.0) · © OpenStreetMap

Built by Team Kanban

31

intakes watched: 8 desalination plants, 22 power-plant cooling intakes, 1 industrial intake.

ACT NOW · Kalba Power Plant
Unusual water 12.1 km to the ENE · 100th percentile vs 33 past same-season images · Sentinel-3 agrees · drift scenario 9.8 km at +6 h · request water samples.

Status rules are fixed and shown on screen. Alerts go out as JSON to the operator's webhook, then email, SMS or the control room.

BLUEBAN 813 · Team Kanban · United Arab Emirates

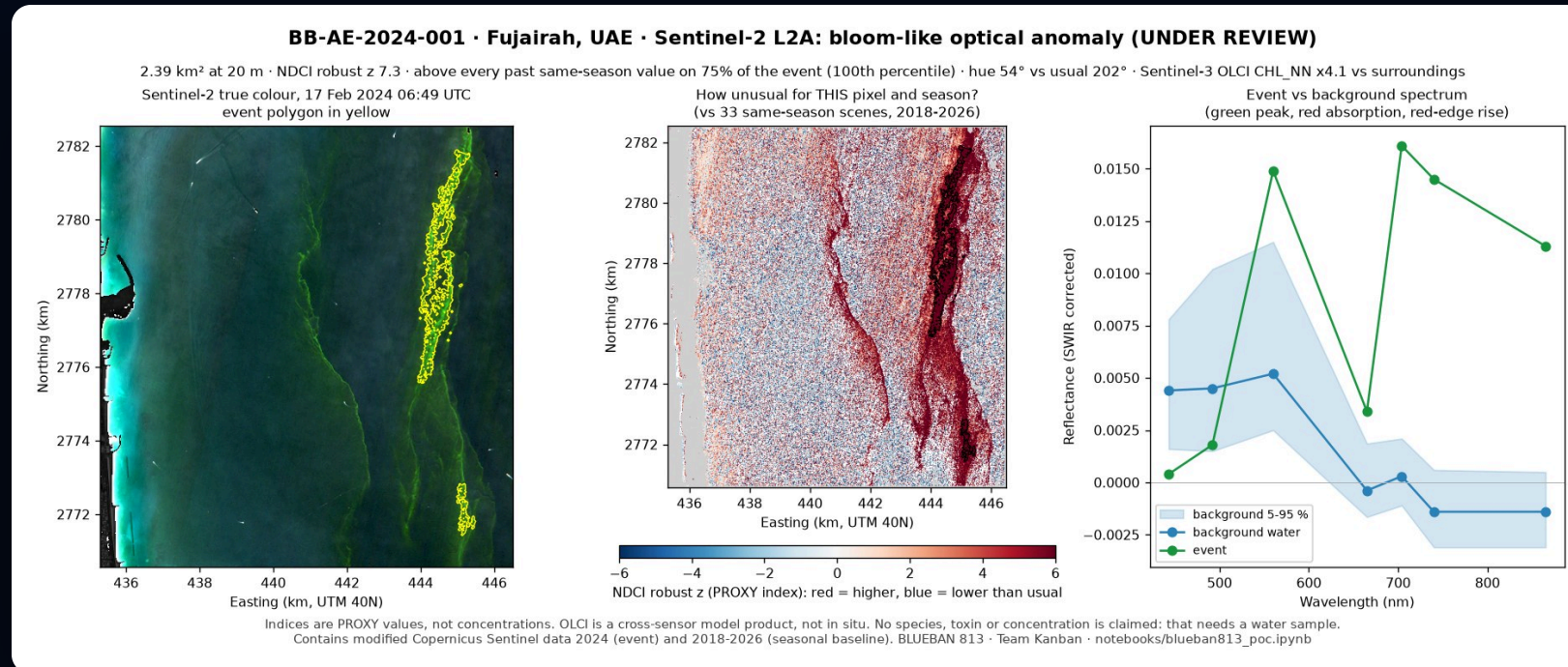
Arab Youth Space Hackathon 2026 · 813 Challenge · 08

Fujairah, 17 February 2024: unusual for this place and season.

z = 7.3 chlorophyll index vs 33 same-season images from other years: 100th percentile

54° vs 202° water colour (hue) vs its usual blue: visibly discoloured

×4.1 Sentinel-3 chlorophyll product, same morning (a model, not a sample)



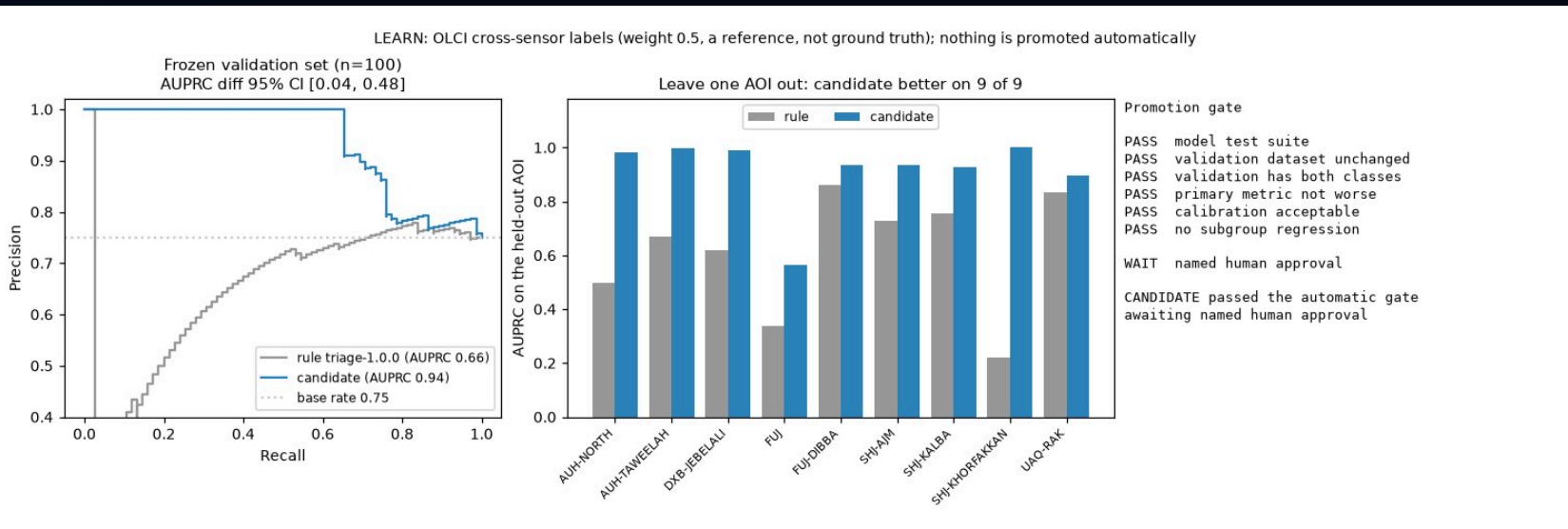
Sentinel-2A, 06:49 UTC · 2.39 km², 8 km offshore · rerun from [notebooks/blueban813_poc.ipynb](https://notebooks.azure.com/blueban813/poc.ipynb) on the committed sample input. No species, toxin or concentration is claimed without a water sample.

The learned model beats the rule, **and still waits for a person.**

0.66 → 0.94 AUPRC on 100 frozen test labels; gain CI [0.04, 0.48]

9 of 9 held-out areas where the candidate beats the rule

5,080 Sentinel-2 images screened over 10 UAE areas



• WHERE IT BREAKS, STATED UP FRONT

- No public UAE water samples: indices stay proxies until lab results calibrate them.
- It stands down on known false alarms: an index spike over unchanged blue water (Sentinel-3 $\times 1.1$) and the Gulf of Annaba.
- Labels come from Sentinel-3, a model product (weight 0.5), not field truth.
- Bloom-like flags agree weakly with Sentinel-3 ($\rho -0.19$, $n = 83$); the drift is a wind-only scenario.

What does 813 add? **Measured, not assumed.**

159 → 84

false alarms on borderline cases at the same catch rate:
Sentinel-2 bands vs simulated 813 bands on the same 25,513
real Tanager pixels (173 spatial blocks).

no gain

on obvious plumes, where both are near-perfect. 813 is the
upgrade for hard, early cases.

196 / 205

simulated 813 bands with real EnMAP support over Shawka
Dam, inland.



Reference: a full-spectrum (368-band) anomaly detector, not field data. Every 813 value is simulated and labelled so; the reader for real 813 products is already written and tested.

Fewer wasted boat trips, earlier decisions, **a record a regulator can trust.**

Impact

- Earlier warning for potable-water and cooling intakes on both UAE coasts.
- Scarce sampling goes where it resolves something, with a clean-water control.
- SDG 6 (6.3, 6.5) and SDG 14 (14.1); ports from the Gulf to the Maghreb without re-tuning, and inland.

Business model

- **Who pays:** desalination and power operators, regulators and municipalities, ports, aquaculture, tourism.
- **Offer:** per-area monitoring · per-asset alerts · per-event 813 hyperspectral analysis · reports and API.
- **Unit economics:** open data in, near-zero hosting; marginal cost per area is compute and storage. No revenue claimed yet.

• WHY BLUEBAN, NOT ANOTHER DASHBOARD

Index maps and satellite viewers



A picture with no baseline: a turbid harbour looks alarming every clear day. BLUEBAN judges each pixel against its own past.

Scheduled boat sampling



Accurate, but one point on a fixed date. BLUEBAN decides where and when the next sample resolves something.

Black-box detection models



BLUEBAN shows its evidence, logs every decision, and promotes a model only after a frozen test and a person's sign-off.

Next: **one coast, one operator, one season.**

1 · **DEPLOY ON GIQ**

The screening pipeline and operator screens, next to the data GIQ hosts.



2 · **REAL 813 DATA**

Swap the simulator for the real 813 reader the day products arrive.



3 · **ONE PILOT**

Compare our flags with one operator's lab record for a season; calibrate concentrations from their samples.

We are asking for: a pilot partner on the UAE coast · access to Satellite 813 data · an incubation seat.